



School of Computer Science Engineering and Technology

Course - B.Tech.
Course Code – CSET-225
Year - 2026

Semester – 5th
Course Name - IMDAI
Semester - ODD

Max. Marks: 3

LAB ASSIGNMENT # 03

Next Word Prediction using RNN, LSTM or GRU.

Objective:

The objective of this assignment is to design, implement, and analyze a **Next Word Prediction system** using **RNN, LSTM (Long Short-Term Memory) and GRU networks**. Students will gain hands-on experience with text preprocessing, sequence modeling, and deep learning-based language modeling.

Dataset

Use the **Shakespeare Hamlet dataset** from the NLTK Gutenberg Corpus.

Tasks

Task 1: Data Collection & Understanding

- Load the dataset using NLTK.
- Display:
 - Total number of words
 - Sample text (first 10 lines)
- Explain:
 - Why this dataset is suitable for language modeling?

Task 2: Text Preprocessing

- Perform the following:
 - Convert text to lowercase
 - Remove unnecessary characters (if any)
 - Tokenization using Keras Tokenizer
- Generate:
 - Word index dictionary
- Explain:
 - Importance of tokenization in NLP

Task 3: Sequence Generation

- Create input sequences using n-gram approach:
 - Example:
"I am learning deep learning"
→ ["I am", "I am learning", "I am learning deep", ...]
 - Convert sequences into numerical format
 - Explain:
 - Why sequence generation is important for next-word prediction?
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Task 4: Padding & Feature Engineering

- Apply padding using `pad_sequences`
 - Separate:
 - Predictors (X)
 - Labels (y)
 - Convert labels into categorical format
 - Explain:
 - Why padding is required?
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Task 5: Model Design using RNN or LSTM or GRU,

- Build a deep learning model using:
 - Embedding Layer
 - LSTM Layer(s)
 - Dense Output Layer with Softmax
 - Compile the model with:
 - Loss: categorical_crossentropy
 - Optimizer: Adam
 - Print model summary
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Task 6: Model Training & Evaluation

- Train the model
 - Plot:
 - Training loss vs epochs
 - Evaluate model performance
 - Discuss:
 - Overfitting / Underfitting
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Task 7: Next Word Prediction

- Write a function:
 - Input: seed text
 - Output: predicted next word
 - Test with at least 5 different inputs
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Task 8: Analysis & Improvements

Answer the following:

1. What are the limitations of LSTM in language modeling?
 2. How can performance be improved?
 - (Hint: Bidirectional LSTM, Transformers, more data, etc.)
 3. Compare LSTM with Transformer Architecture (briefly)
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Bonus Task (Optional)

- Implement **Bidirectional LSTM** OR
 - **Compare results of all the models.**
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Submission Guidelines

- Submit:
 - Python Notebook (.ipynb)
 - Summary Report
- Report must include:
 - Methodology
 - Results
 - Graphs
 - Observations